Case Study on the Analysis of SDLC Models Applied to an E-Commerce Platform

AIM

To analyze the Software Development Life Cycle (SDLC) models used in the development of an E-Commerce Platform for an Online Retail Business and identify the most suitable SDLC model for building a secure, scalable, and user-friendly online shopping application.

OBJECTIVES

- Study different Software Development Life Cycle (SDLC) models.
- Analyze the software development process of an E-Commerce Platform.
- Identify the functional and non-functional requirements of the system.
- Compare the advantages and limitations of various SDLC models.
- Recommend the most appropriate SDLC model for the application.
- Understand how Agile and DevOps practices improve software quality, delivery speed, and customer satisfaction.

SELECTED SOFTWARE APPLICATION

E-Commerce Platform for an Online Retail Business

Overview

An E-Commerce Platform enables customers to purchase products online from anywhere using a web or mobile application. It provides a convenient and secure shopping experience by allowing users to browse products, compare prices, place orders, make online payments, and track deliveries in real time.

The platform also provides administrators with tools to manage products, inventory, customer information, orders, payments, and business reports from a centralized dashboard. The application automates business operations, improves efficiency, and enhances customer satisfaction.

PROBLEM STATEMENT

ABC Retail is a growing retail company that currently relies on physical stores and manual business operations. Customers are required to visit stores or place orders through phone calls, making the purchasing process slow and inconvenient. Product inventory is maintained manually, resulting in stock inaccuracies, delayed order processing, billing errors, and poor inventory management.

The company also lacks a centralized system to manage customer information, orders, payments, and sales reports. Customers cannot track their orders after purchase, while management has limited visibility into business performance.

With the increasing demand for online shopping, these challenges reduce customer satisfaction, increase operational costs, and limit business growth. Therefore, the company requires a modern E-Commerce Platform that automates business processes and delivers a seamless online shopping experience.

SOLUTION

The proposed solution is to develop a secure, scalable, and user-friendly E-Commerce Platform that automates the complete online retail process.

The application will allow customers to:

- Register and log in securely
- Browse and search products
- Filter products by category, price, or brand
- Add products to a shopping cart or wishlist
- Place orders
- Make secure online payments
- Track order status in real time
- View purchase history
- Submit ratings and reviews

The administrator will be able to:

- Manage products and categories
- Monitor inventory
- Process customer orders
- Manage users
- Configure discounts and offers
- Generate sales and inventory reports
- Monitor business performance through dashboards

The system will integrate secure payment gateways, inventory management, and automated notifications to improve operational efficiency. Agile SDLC with DevOps practices will be adopted to ensure continuous development, testing, deployment, and customer feedback.

CONCLUSION

The case study concludes that the Agile SDLC model is the most suitable approach for developing an E-Commerce Platform because it supports continuous improvements, quick feature delivery, and changing customer requirements. When combined with DevOps practices, it ensures efficient development, reliable deployment, and high-quality software, resulting in improved customer satisfaction and business growth.